

# ECO310 Midterm 1 - Readable Cheat-sheet

## 1 Preferences & Axioms

### 1.1 Rational Preferences

**Complete:** Can compare any two bundles ( $\forall x, y: x \succeq y$  or  $y \succeq x$ )

**Transitive:** If  $x \succeq y$  and  $y \succeq z$ , then  $x \succeq z$

**Monotonicity:** More is better  $\Rightarrow$  strictly prefer more of any good

**Continuity (LNS):** No sudden jumps. For any  $x$  and  $\varepsilon > 0$ , exists  $y$  very close to  $x$  where  $y \succ x$

**Convexity:** Mixtures  $\geq$  extremes. Upper contour sets are convex. (Means diminishing MRS)

### 1.2 Utility Functions

**Ordinal:** Only ranking matters, not magnitude

**MRS:**  $-\frac{dx_2}{dx_1}\Big|_{IC} = \frac{MU_1}{MU_2}$  (Slope of IC, bang-per-buck)

At optimum:  $MRS = \frac{p_1}{p_2}$

### 1.3 Common Utility Forms

**Cobb-Douglas:**  $u = x_1^a x_2^b$

- $MRS = \frac{a}{b} \cdot \frac{x_2}{x_1}$
- Demand:  $x_1^* = \frac{a}{a+b} \cdot \frac{m}{p_1}$ ,  $x_2^* = \frac{b}{a+b} \cdot \frac{m}{p_2}$
- **Magic rule:** Spend fraction  $\frac{a}{a+b}$  on good 1

**Perfect Complements:**  $u = \min\{ax_1, bx_2\}$

- Always consume at kink:  $ax_1 = bx_2$
- $MRS = 0$  on horizontal,  $-\infty$  on vertical
- Demand:  $x_1^* = \frac{m}{bp_1 + ap_2}$ ,  $x_2^* = \frac{am}{bp_1 + ap_2}$
- **No substitution effect** (entire change is income effect)

**Perfect Substitutes:**  $u = ax_1 + bx_2$

- $MRS = -a/b$  (constant)
- Buy all of cheaper good (per unit of utility)
- If  $\frac{a}{p_1} > \frac{b}{p_2}$ : buy only good 1
- If  $\frac{a}{p_1} < \frac{b}{p_2}$ : buy only good 2

**Quasilinear:**  $u = v(x_1) + x_2$

- $MRS = v'(x_1)$  (depends only on  $x_1$ )
- ICs are vertical translates
- **No income effect** for  $x_1$  (at interior)

**CES:**  $u = (x_1^r + x_2^r)^{1/r}$

- $MRS = -(x_2/x_1)^{1-r}$
- $r \rightarrow 0$ : Cobb-Douglas
- $r \rightarrow -\infty$ : Perfect complements
- $r = 1$ : Perfect substitutes

## 2 Consumer Choice (UMP)

### 2.1 Budget Constraint

$$p_1 x_1 + p_2 x_2 \leq m$$

**Slope:**  $-p_1/p_2$

**Intercepts:**  $(m/p_1, 0)$  and  $(0, m/p_2)$

### 2.2 Interior Optimum

**Tangency condition:**  $\frac{MU_1}{p_1} = \frac{MU_2}{p_2}$

Equivalently:  $MRS = \frac{p_1}{p_2}$

**Budget binds:**  $p_1 x_1 + p_2 x_2 = m$

### 2.3 Lagrangian Method

$$\mathcal{L} = u(x_1, x_2) + \lambda[m - p_1 x_1 - p_2 x_2]$$

**FOCs:**  $MU_i = \lambda p_i$  for  $i = 1, 2$

$\lambda$  = marginal utility of income

### 2.4 Corner Solutions

If  $MRS > p_1/p_2$  everywhere: consume only  $x_1$

If  $MRS < p_1/p_2$  everywhere: consume only  $x_2$

**Check:** Always verify  $x_i \geq 0$ !

### 2.5 Demand Properties

**Marshallian demand:**  $x_i(p, m)$

- Homogeneous degree 0 in  $(p, m)$
- Decreasing in  $p_i$ , increasing in  $m$  (for normal goods)

**Indirect utility:**  $v(p, m) = u(x^*(p, m))$

- Decreasing in  $p_i$ , increasing in  $m$

## 3 Expenditure & Duality

### 3.1 Hicksian Demand

**Hicksian demand:**  $h_i(p, \bar{u})$

Minimizes expenditure to reach utility  $\bar{u}$  at prices  $p$

**Shepard's Lemma:**  $h_i = \frac{\partial e}{\partial p_i}$

### 3.2 Expenditure Function

$$e(p, \bar{u}) = p \cdot h(p, \bar{u})$$

Minimum cost to reach utility  $\bar{u}$

### 3.3 Duality Relations

$$x_i(p, m) = h_i(p, v(p, m))$$

$$h_i(p, \bar{u}) = x_i(p, e(p, \bar{u}))$$

$$v(p, e(p, \bar{u})) = \bar{u}$$

$$e(p, v(p, m)) = m$$

## 4 Income & Substitution Effects

### 4.1 Slutsky Decomposition

**Total effect** = Substitution effect + Income effect

When  $p_1$  rises:

**Substitution effect:** Always  $\leq 0$  (buy less)

- Move along IC to new price ratio
- Compute: compensated bundle - original bundle

**Income effect:** Depends on normal/inferior

- Normal good:  $< 0$  (buy less)
- Inferior good:  $> 0$  (buy more)
- Compute: new bundle - compensated bundle

**Giffen good:** Income effect dominates  $\Rightarrow$  demand increases when price rises

### 4.2 Recipe for Calculations

1. Find **original bundle** at initial prices
2. Find **compensated bundle** (new prices, old utility)
3. Sub effect = comp - orig
4. Income effect = new - comp

## 5 Choice Under Uncertainty

### 5.1 Expected Utility (vNM)

$$EU(L) = \sum_i p_i u(c_i)$$

**Axioms:** Completeness, Transitivity, Continuity, Independence

**vNM utility unique up to positive affine transforms:**

$$\tilde{u} = a + bu, \quad b > 0$$

(NOT arbitrary monotone transforms!)

### 5.2 Risk Attitudes

**Risk averse**  $\Leftrightarrow u$  concave  $\Leftrightarrow u'' < 0$

- Jensen's inequality:  $u(E[x]) > E[u(x)]$
- Prefer certainty to gambles

**Risk neutral**  $\Leftrightarrow u$  linear

**Risk loving**  $\Leftrightarrow u$  convex  $\Leftrightarrow u'' > 0$

## 5.3 Risk Premium & Certainty Equivalent

**CE:** Amount for certain worth same as gamble

$$u(CE) = EU$$

**Risk Premium:**  $RP = E[x] - CE$

- Risk averse:  $RP > 0$  (pay to avoid risk)
- Risk neutral:  $RP = 0$
- Approximation:  $RP \approx \frac{1}{2} r_A(\bar{x}) \cdot \sigma^2$

## 5.4 Arrow-Pratt Measures

**Absolute risk aversion:**

$$r_A(x) = -\frac{u''(x)}{u'(x)}$$

**Relative risk aversion:**

$$r_R(x) = x \cdot r_A(x) = -\frac{xu''(x)}{u'(x)}$$

Higher  $r_A \Rightarrow$  higher risk premium, more risk averse

## 5.5 Common Utility Functions

**CRRA:**  $u = \frac{x^{1-\rho}}{1-\rho}$  or  $u = \ln x$  (if  $\rho = 1$ )

- $r_A = \rho/x$  (DARA)
- $r_R = \rho$  (constant)

**CARA:**  $u = -e^{-\alpha x}$

- $r_A = \alpha$  (constant)
- $r_R = \alpha x$  (IRRA)

**Log:**  $u = \ln x$

- $r_A = 1/x$  (DARA)
- $r_R = 1$  (constant)

**DARA:** Decreasing absolute risk aversion  $\Rightarrow r'_A < 0$  (more wealth  $\Rightarrow$  willing to take larger fixed-\$ gambles)

## 5.6 Stochastic Dominance

**FOSD:**  $F(x) \leq G(x)$  for all  $x$

- All  $u' > 0$  prefer  $F$

**SOSD:**  $\int_{-\infty}^t F(x)dx \leq \int_{-\infty}^t G(x)dx$  for all  $t$

- All risk-averse prefer  $F$
- Same mean:  $G$  is mean-preserving spread of  $F$

## 6 Insurance & Portfolio

### 6.1 Insurance Problem

**Setup:** Wealth  $w$ , loss  $D$  with probability  $\pi$

Premium  $q$  per dollar, payout  $K$

$$EU = (1 - \pi)u(w - q) + \pi u(w - q - D + K)$$

**Fair insurance:**  $q = \pi K$

- Risk-averse  $\Rightarrow$  full insurance ( $K = D$ )
- Consumption same in both states

**Unfair insurance:**  $q > \pi K$

- Partial insurance ( $K < D$ )

**State-contingent view:**

$$MRS_{12} = \frac{\pi_1 u'(c_2)}{\pi_2 u'(c_1)}$$

On 45° line ( $c_1 = c_2$ ):  $MRS = \pi_1/\pi_2$

Fair prices ( $q_s = \pi_s$ ): choose  $c_1 = c_2$

### 6.2 Portfolio Choice

Invest  $a$  in risky (return  $\tilde{r}$ ), rest at  $r_f$

$$\tilde{W} = w(1 + r_f) + a(\tilde{r} - r_f)$$

**FOC:**  $E[u'(\tilde{W})(\tilde{r} - r_f)] = 0$

If  $E[\tilde{r}] > r_f$ : risk-averse invests  $a^* > 0$

**Portfolio std dev:**  $\sigma_P = \alpha \cdot \sigma_{\text{risky}}$

where  $\alpha = a/w$  (fraction in risky)

**Expected return:**  $\alpha E[\tilde{r}] + (1 - \alpha)r_f$

## 7 Markets & General Equilibrium

### 7.1 Edgeworth Box

2 consumers, 2 goods

Dimensions: total endowments  $E_x, E_y$

**Feasible:**  $x^1 + x^2 = E_x, y^1 + y^2 = E_y$

### 7.2 Pareto Efficiency

No Pareto improvement exists

**Interior PE:**  $MRS^1 = MRS^2$  at all PE allocations

### 7.3 Competitive Equilibrium

**Each consumer:** Maximize utility s.t. budget  $p \cdot x = p \cdot e$

**Markets clear:**  $\sum \text{demands} = \sum \text{endowments}$

**Walras' Law:**  $p \cdot ED_x + p \cdot ED_y = 0$

- If  $n - 1$  markets clear,  $n$ -th clears
- Only relative prices matter

### 7.4 Welfare Theorems

**1st:** CE  $\Rightarrow$  Pareto efficient (needs LNS)

**2nd:** Any PE can be CE with transfers (needs convexity)

### 7.5 Solving for CE

1. Set  $p_2 = 1$  (numeraire)

2. Solve each consumer's problem: get  $x^i(p_1)$

3. Market clearing:  $\sum x_1^i(p_1) = E_x$

4. Solve for  $p_1^*$

5. Check  $x_2^i \geq 0$  for all

## 8 WARP & Revealed Preference

### 8.1 Weak Axiom (WARP)

If  $x$  chosen when  $y$  affordable, and  $y$  chosen in period  $B$ , then  $y$  must NOT be affordable when  $x$  was chosen

**Check:** For two periods  $(p^A, m^A)$  and  $(p^B, m^B)$ :

1. Cost of A at B's prices vs B's cost?
2. Cost of B at A's prices vs A's cost?
3. Both strictly cheaper  $\Rightarrow$  WARP violation

### 8.2 Sen's Property

$\alpha$ : No new choices from smaller sets

$\beta$ : Choice from subset  $\Rightarrow$  choice from whole set

$\alpha + \beta \Leftrightarrow$  Rationalizable  $\Leftrightarrow$  WARP

## 9 Problem-Solving Tips

### 9.1 General Recipe

1. **Identify states** and probabilities
2. **Write EU** as  $\sum p_s u(w_s)$
3. **Take FOC** w.r.t. choice variable
4. **Solve** for optimal choice
5. **Check corners** ( $x = 0$  or  $x = w$ )

### 9.2 Key Checks

- Always verify  $x_i \geq 0$
- Perfect complements: sub effect = 0
- CD exponents: normalize  $\alpha = a/(a + b)$
- vNM utility: only affine transforms
- Portfolio:  $\sigma^2$  (variance)  $\neq \sigma$  (std dev)
- Corner: If FOC gives  $x < 0$ , set  $x = 0$
- "Spend  $\times$ "  $\neq$  "Buy  $\times$ " (unless prices equal)
- Risk-averse prefers any mean-preserving contraction

### 9.3 Common Traps

- Don't confuse UMP ( $x(p, m)$ ) vs EMP ( $h(p, \bar{u})$ )
- MRS direction: be consistent with sign
- Convexity: show  $|MRS|$  decreasing along IC
- WARP: "revealed preferred" = chosen when other affordable
- EU states: follow probability tree to final wealth
- Quasilinear interior  $x_1$  independent of  $m$ , but check  $x_2 \geq 0$
- \$90 certain vs 50-50 \$0/\$200 is ambiguous for risk-averse